



NexusHR

AI-Enabled Enterprise HR & Workforce Intelligence Platform

Employee Lifecycle · Attendance · Payroll · Performance · AI Insights

99.95%

Uptime SLA

<300ms

API Latency

50k+

Employees

40–60%

Admin Saved

PROJECT CODE AMX-JFS-2026-04

VERSION 1.0 – Industry Edition

DOMAIN Java Full-Stack

DATE April 2026

PREPARED FOR Amdox – Engineering Division

CLASSIFICATION Internal

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Executive Overview

Mission statement

NexusHR is a production-grade Java full-stack HR management platform covering the complete employee lifecycle — from onboarding to offboarding — with real-time attendance, automated payroll, 360-degree performance reviews, and AI-driven workforce intelligence. Designed for mid-to-large enterprises (5,000–50,000 employees), it eliminates manual spreadsheets, reduces HR operational costs by 40–60%, and provides real-time decision-making insights to HR leaders.

Why NexusHR matters for Amdox Technologies

Modern enterprises in manufacturing, IT services, banking, and retail need integrated HR systems that scale seamlessly. NexusHR positions Amdox as an enterprise-grade full-stack partner, delivering 40–60% HR workload automation, 15–25% improvement in employee retention through AI insights, and a zero-downtime, multi-tenant deployment model that can be productized for Amdox's enterprise clients.

1.1 Business impact targets

Business Objective	Metric	Target	Measurement
HR Workload Reduction	Manual process hours	↓ 40–60%	Pre/post deployment survey
Employee Retention	30-day attrition rate	↑ 15–25%	AI attrition model accuracy
Payroll Accuracy	Payroll error rate	< 0.1%	Audit of monthly payroll runs
System Uptime	Monthly availability	99.95%	Uptime monitoring

Business Objective	Metric	Target	Measurement
API Performance	P95 response time	< 300ms	Prometheus latency metric
Concurrent Users	Peak HR users (payroll cycle)	10,000+	k6 load test
Deployment Frequency	Release cadence	Zero-downtime weekly	Blue-green deployment log

1.2 Target users & use cases

User Role	Primary Use Cases	Access Level
HR Administrators	Employee CRUD, payroll runs, compliance reports, org chart management	Full platform access
HR Managers	Department analytics, leave approvals, performance review oversight	Department-scoped
Line Managers	Team attendance, goal setting, 360° feedback initiation, talent reports	Team-scoped
Employees	Self-service: leave requests, payslip download, performance goals, profile	Self-service only
Finance Teams	Payroll reports, tax summaries, cost-centre analytics, budget forecasts	Finance module only
C-Suite / Executives	Workforce dashboards, attrition predictions, headcount planning	Read-only analytics
IT Administrators	SSO configuration, audit logs, security policies, role management	System configuration

1.3 Non-functional requirements

NFR	Requirement	Target	Validation
Availability	Monthly uptime SLA	99.95%	UptimeRobot + PagerDuty
API Latency	P95 response time for core endpoints	< 300ms	Prometheus + Grafana
Dashboard Load	Initial page load (React SSR)	< 2s	Lighthouse performance
Throughput	Concurrent HR users during payroll	10,000+	k6 load testing
Scalability	Auto-scale on CPU > 70%	Horizontal Kubernetes HPA	HPA stress test
Security	OWASP Top 10 2021 compliance	Full mitigation	OWASP ZAP scan + Snyk
Data Encryption	Sensitive fields at rest	AES-256 + TLS 1.3	Security audit
Audit Trail	All sensitive mutations logged	100% coverage	Immutable audit log review

2 Core Functional Requirements

NexusHR delivers ten production-grade modules covering the complete HR value chain. Each module includes granular acceptance criteria and integration touchpoints across the platform.

ID	Module	Detailed Description & Business Value	Acceptance Criteria
F-01	Employee Lifecycle Management	Complete onboarding-to-offboarding workflow: digital offer letters, document upload (S3), profile management, org chart (recursive tree), role assignment, transfer workflow, exit interview, and offboarding checklist with approval steps.	Onboarding completes in < 5 steps; document upload to S3 in < 3s; org chart renders up to 10 hierarchy levels; offboarding triggers automated access revocation
F-02	Attendance & Leave Management	Biometric simulation API (clock-in/out), geo-fence validation, shift scheduling, leave type configuration (casual/sick/earned/maternity), multi-level approval workflow, leave balance accrual engine with carry-forward rules.	Real-time attendance dashboard < 1s refresh; leave balance accurate to day; approval notification within 30s via WebSocket; overtime calculation per labour rules
F-03	Payroll Processing Engine	Configurable gross-to-net payroll: salary components, tax slab engine (regime-aware), PF/ESI/TDS deductions, CTC breakdown, payslip PDF generation (iText), bulk payroll run with parallel processing (Java virtual threads), payroll journal export.	Payroll run for 10k employees < 5 min; payslip PDF generated in < 2s; zero rounding errors; export to Excel/PDF; full audit trail per payroll cycle
F-04	Performance Management	Goal-setting (OKR framework), mid-year and annual review cycles, 360-degree feedback (peer, manager, subordinate), calibration sessions, bell-curve normalization, performance scorecard, succession planning flags.	Review cycle completion tracked in real-time; SHAP-explained AI rating suggestions; calibration dashboard with forced distribution; export performance reports
F-05	AI Workforce Intelligence	Spring AI + OpenAI/Ollama integration: predictive attrition model (Random Forest), skill gap analysis (NLP on JD vs profile), engagement scoring (survey + activity signals), headcount forecasting, AI-generated retention action recommendations.	Attrition model AUC ≥ 0.82 ; skill gap report per employee; engagement score updated weekly; recommendations explained in plain English; model retrain monthly
F-06	Recruitment & Onboarding	Job requisition, ATS integration (webhook), candidate pipeline, interview scheduling, offer management, background check status, digital onboarding kit with e-signature (DocuSign API).	Requisition to offer in tracked stages; webhook delivery < 5s; onboarding kit auto-generated on offer accept; e-signature status synced in real-time
F-07	Admin & Manager Dashboards	Role-based React dashboards: headcount pyramid, attrition waterfall, cost-per-hire, absenteeism heatmap, payroll cost trend, department performance radar, real-time workforce KPIs with drill-down to employee level.	Dashboard loads < 2s; charts render with Recharts + D3; export to PDF/Excel; dark mode; mobile responsive at 375px; WCAG 2.1 AA accessible
F-08	Notification & Communication	Multi-channel notification engine: in-app (WebSocket), email (Spring Mail + AWS SES), SMS (Twilio), push (FCM). Event-driven via Kafka: leave approved, payslip ready, appraisal due, policy update.	Notification delivery < 30s P95; email success rate $\geq 99\%$; Kafka consumer lag < 100ms; user notification preferences respected; unsubscribe honoured
F-09	Compliance & Audit Engine	GDPR-aligned data subject requests, statutory report generation (Form 16, EPF ECR, ESI), labour law compliance checklist, immutable audit log (append-only Postgres table with hash chaining), data retention policy enforcement.	DSR fulfilled < 72h; Form 16 PDF generated in < 10s; audit log tamper-evident; retention policy auto-archives data after configured period
F-10	Reporting & Analytics	Self-service report builder: drag-and-drop field selector, filter/group/sort, scheduled email delivery, custom dashboard widgets, export (CSV/Excel/PDF), embedded BI charts, API data export for BI tools.	Report builder saves config in < 500ms; scheduled delivery via Quartz scheduler; export < 30s for 50k rows; API export supports pagination + filtering

3 Production Technology Stack – 2026

Every technology was evaluated against enterprise-grade criteria: performance under load, security posture, ecosystem maturity, and long-term support. The stack is optimized for a 5,000–50,000 employee deployment running on AWS EKS.

Layer	Technology	Version	Rationale / Alternatives
Backend Runtime	Java 21 LTS	21	Virtual threads (Project Loom) for massive concurrency; GraalVM native considered
Backend Framework	Spring Boot 3.3	3.3	Production battle-tested; Spring ecosystem depth; Quarkus evaluated
API Design	Spring MVC REST + WebSocket	6.x	OpenAPI 3.1 spec; WebSocket for real-time; gRPC for service-to-service
Security	Spring Security 6 + JWT (RS256)	6.x	RBAC + ABAC; Argon2id password hashing; OAuth2 + OIDC (Keycloak)
ORM / DB Access	Spring Data JPA + Hibernate 6	3.x	JPQL + native queries; N+1 prevention; QueryDSL for dynamic queries
Database	PostgreSQL 17	17	ACID; JSONB for flexible fields; partitioned audit tables; TimescaleDB extension
DB Migrations	Flyway	10.x	Repeatable migrations; baseline on existing schemas; Liquibase evaluated
Caching	Redis 7+ (Redisson)	7.x	Session store; distributed locks; pub/sub; write-through cache strategy
Messaging	Apache Kafka	3.7	Event-driven architecture; audit log streaming; RabbitMQ evaluated
Search	Elasticsearch 8.15	8.15	Employee search; document indexing; Spring Data Elasticsearch
File Storage	AWS S3 + CloudFront	—	Documents, payslips, profile photos; pre-signed URLs; 7-year retention
PDF Generation	iText 7 / Apache PDFBox	7.x	Payslips, Form 16, offer letters; digital signature embedding
Email	Spring Mail + AWS SES	—	Transactional; DKIM/SPF; bounce handling; rate-limited queuing
SMS	Twilio API	—	OTP, leave alerts; Textlocal as Indian fallback
AI Integration	Spring AI 1.0 + OpenAI / Ollama	1.0	LLM orchestration; RAG for HR policy Q&A; HuggingFace for on-prem models
Frontend Framework	React 19 + TypeScript 5.5	19	Concurrent mode; server components; Next.js App Router evaluated
UI Components	shadcn/ui + Radix + Tailwind 4	—	Accessible; unstyled primitives; MUI too opinionated for custom brand
State Management	Zustand + TanStack Query v5	—	Lightweight; TanStack for server state with cache invalidation
Charts	Recharts + D3.js	—	Recharts for dashboards; D3 for custom org chart + heatmaps
Build Tool	Vite 6	6.x	HMR; tree-shaking; Rollup bundler; Webpack 5 evaluated
Containerization	Docker 27 (Distroless base)	27	Multi-stage; minimal attack surface; non-root user; CAP_DROP ALL
Orchestration	Kubernetes 1.31 + Helm 3	1.31	EKS deployment; HPA + VPA; PodDisruptionBudget; ArgoCD GitOps

Layer	Technology	Version	Rationale / Alternatives
Service Mesh	Istio 1.22	1.22	mTLS between services; traffic management; circuit breaker
CI/CD	GitHub Actions + ArgoCD	—	Matrix: lint → test → build → push → deploy; manual gate for prod
Monitoring	Prometheus + Grafana + Loki	—	Metrics + dashboards + logs; OpenTelemetry traces; Sentry errors
Security Scanning	OWASP ZAP + Snyk + Trivy	—	DAST + SCA + container scanning in CI pipeline
Infrastructure	Terraform 1.9 + Terragrunt	1.9	AWS EKS + RDS + ElastiCache; Atlantis for GitOps IaC
Testing	JUnit 5 + Mockito + Testcontainers	—	Unit + integration + contract; Playwright E2E; k6 load testing

4 Architecture Overview

4.1 Full-stack system architecture

NexusHR follows a layered microservices architecture with a React SPA client, Spring Cloud Gateway, domain-specific Spring Boot microservices, a polyglot data layer, and a comprehensive observability stack. All inter-service communication is secured via Istio mTLS.

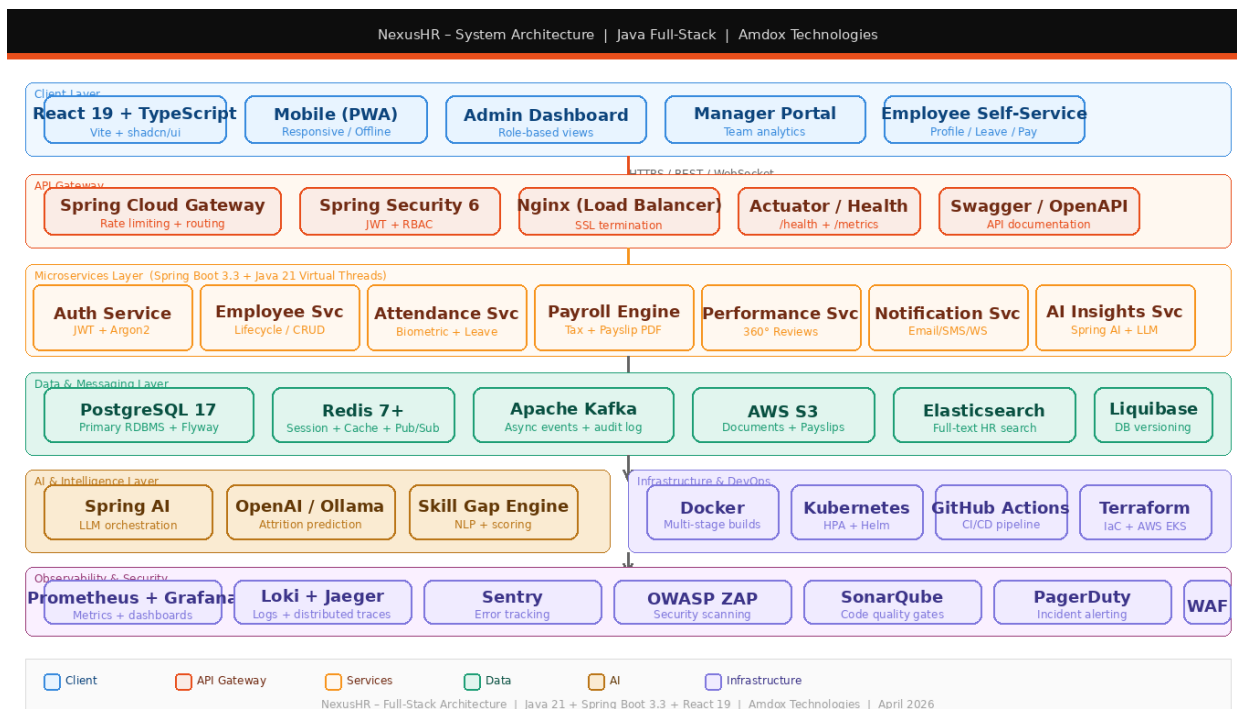


Figure 4.1 – NexusHR Full-Stack System Architecture | Java 21 + React 19 | Amdox Technologies | April 2026

4.2 Request lifecycle & module flow

Every user action travels through a consistent path: React frontend → Nginx load balancer → Spring Cloud Gateway → Spring Security (JWT validation + RBAC) → domain service → data layer → Kafka event bus → AI engine where applicable. The response path is symmetric with Redis caching at the service layer.

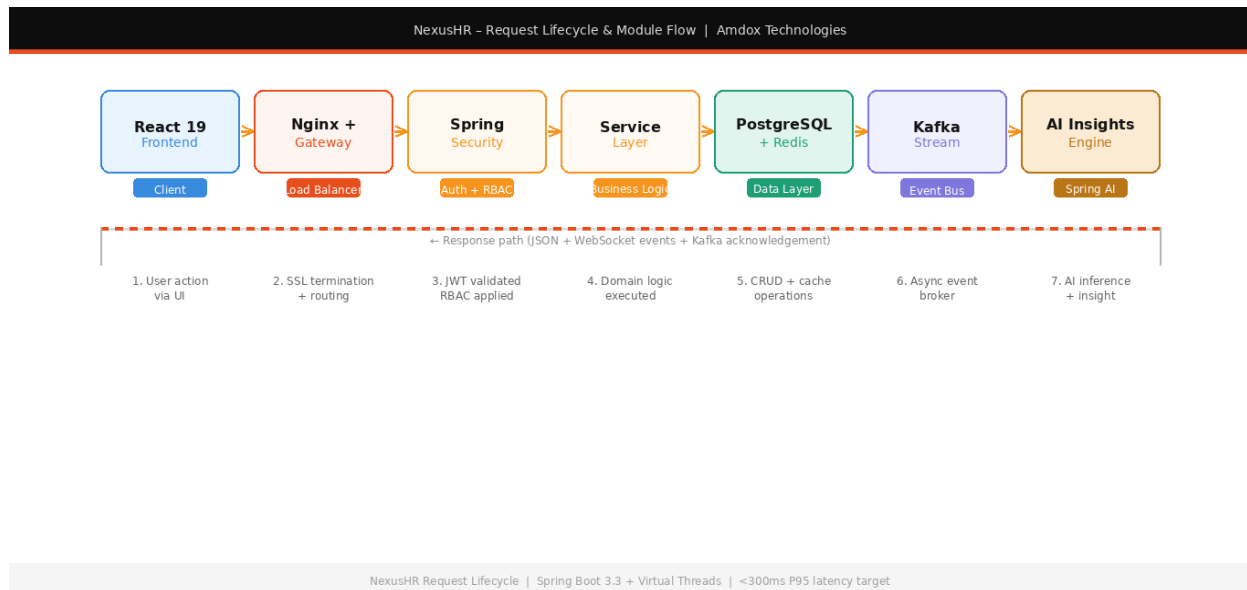


Figure 4.2 – NexusHR Request Lifecycle & Module Flow | <300ms P95 Latency | April 2026

4.3 Service boundaries & responsibilities

Service	Tech	Owns	Key Patterns
Auth Service	Spring Boot + Keycloak	JWT issuance, refresh, MFA, SSO	Stateless JWT (RS256); token blacklist in Redis; PKCE flow
Employee Service	Spring Boot + JPA	Employee profiles, org chart, lifecycle state machine	DDD aggregate; CQRS read model; event sourcing for history
Attendance Service	Spring Boot + WebSocket	Clock-in/out, shift engine, leave engine	Real-time via STOMP over WebSocket; Redis sorted-set for shifts
Payroll Engine	Spring Boot + Virtual Threads	Gross-to-net calc, payslip PDF, journal export	Saga pattern; parallel processing with virtual threads; idempotent runs
Performance Service	Spring Boot	OKRs, reviews, calibration, 360° feedback	State machine for review lifecycle; event-driven notifications
AI Insights Service	Spring AI + Python sidecar	Attrition model, skill gap NLP, engagement scoring	Spring AI orchestration; Python FastAPI sidecar for ML inference
Notification Service	Spring Boot + Kafka consumer	Email, SMS, WebSocket, push	At-least-once delivery; idempotent consumer; dead-letter queue

4.4 Data model highlights

Entity / Table	DB	Key Fields	Notes
employees	PostgreSQL	id, tenant_id, status, hire_date, grade, manager_id	Soft-delete; row-level tenant isolation; recursive self-join for org chart
attendance_records	PostgreSQL + TimescaleDB	employee_id, clock_in, clock_out, geo_lat, geo_lng	Time-series partitioning by month; hypertable for fast range queries

Entity / Table	DB	Key Fields	Notes
payroll_runs	PostgreSQL	run_id, period, status, total_gross, total_net, approved_by	Idempotency key per run; immutable once approved; audit trail
performance_reviews	PostgreSQL	review_id, reviewee_id, reviewer_id, cycle, score, stage	State machine (draft→submitted→calibrated→published); JSONB for custom fields
audit_log	PostgreSQL (append-only)	id, actor_id, action, entity, before_json, after_json, hash	Hash-chained rows; trigger-based insertion; never updated or deleted
ai_predictions	PostgreSQL	employee_id, model_version, attrition_score, skill_gaps, generated_at	Versioned by MLflow model tag; refreshed weekly; SHAP values stored as JSONB

5 Granular 28-Day Execution Plan

Week 1 – Core backend, authentication & employee module

Day 1 – Project setup & monorepo architecture

- Maven multi-module project: api-gateway, auth-service, employee-service, payroll-service, attendance-service, performance-service, ai-service, notification-service, common-lib
- Spring Boot 3.3 parent POM with shared dependency management (Java 21, Lombok, MapStruct, OpenAPI)
- Docker Compose local stack: PostgreSQL 17, Redis 7, Kafka 3.7, Elasticsearch 8, Keycloak 25
- Monorepo Git strategy: conventional commits, GitHub branch protection, CODEOWNERS
- GitHub Actions skeleton: lint (Checkstyle + SpotBugs) + unit test matrix

Day 2 – Authentication & Keycloak integration

- Keycloak realm configuration: realm-per-tenant strategy, OIDC client, SAML adapter for enterprise SSO
- Spring Security 6: JWT filter (RS256), Argon2id password encoder, RBAC with @PreAuthorize
- Auth endpoints: /auth/register, /auth/login, /auth/refresh, /auth/logout (Redis token blacklist)
- MFA: TOTP via Google Authenticator integration (TOTP library + QR code generation)
- Unit tests: auth controller (80% coverage target), MockMvc test harness

Day 3 – Employee domain & org chart

- Employee entity + JPA mapping: hire, transfer, promotion, termination lifecycle states
- Flyway migration V1: employees, departments, roles, grades, employment_history tables
- Org chart API: recursive CTE query for hierarchy; Spring Data JPA projections
- Employee CRUD REST endpoints with OpenAPI 3.1 annotations + Swagger UI
- Testcontainers integration test: PostgreSQL container for repository layer tests

Day 4 – PostgreSQL schema + Spring Cloud Gateway

- Complete Flyway schema: all domain tables with indexes, foreign keys, soft-delete columns
- Row-level security strategy: tenant_id injected at query layer via Hibernate filter
- Spring Cloud Gateway: route config for all services, JWT relay filter, rate limiting (Redis)
- Nginx reverse proxy config: SSL termination, upstream health checks, load balancing
- API versioning strategy: URL versioning (/api/v1/) with deprecation header support

Day 5 – Redis caching + attendance module foundation

- Redisson Spring Boot starter: session store, distributed lock for payroll runs, pub/sub
- Cache-aside strategy: employee profile cache with 5-min TTL, manual eviction on update
- Attendance domain: AttendanceRecord entity, clock-in/out endpoints, geo-validation service
- WebSocket setup: STOMP broker relay, attendance dashboard real-time push
- Leave type configuration: casual, sick, earned, maternity, paternity with carry-forward rules

Day 6 – Leave management & approval workflow

- Leave request state machine: PENDING → MANAGER_APPROVED → HR_APPROVED → ACTIVE (Spring StateMachine)
- Leave balance accrual engine: monthly pro-rata calculation with weekend exclusion
- Multi-level approval workflow: configurable levels per leave type, delegation support
- Kafka event: leave.approved / leave.rejected → notification-service consumer
- Remaining balance calculation API with carry-forward and lapse rules

Day 7 – Week 1 checkpoint

- Deliverables: auth service (JWT + MFA), employee CRUD + org chart, attendance + leave module
- Postman collection: all Week 1 endpoints tested + automated collection runner
- API documentation: Swagger UI live at /swagger-ui.html with example payloads
- ADR-001: Multi-tenant row-level security approach documented
- Code review pass: security checklist, N+1 query audit, input validation review

Week 2 – Payroll engine, performance module & React frontend

Day 8 – Payroll calculation engine

- Gross-to-net payroll: configurable salary components (basic, HRA, TA, special allowance)
- Tax engine: Old vs New regime (India), TDS slab calculation with rebate rules
- Statutory deductions: PF (Employee 12% + Employer 12%), ESI (0.75% employee + 3.25% employer)
- Payroll run orchestration: parallel virtual thread processing for 10k employees
- Idempotency: payroll_run_id prevents duplicate processing; Saga for multi-step commit

Day 9 – Payslip PDF generation & payroll reports

- iText 7 payslip template: company letterhead, salary breakdown table, digital stamp
- S3 upload: payslip PDF stored with pre-signed URL (7-day expiry) for employee download
- Payroll journal export: Excel (Apache POI) for finance team, EPF ECR format generation
- Form 16 Part A generation: TDS certificate PDF with employer digital signature placeholder
- Bulk payslip email: Kafka producer → notification-service → AWS SES batch send

Day 10 – Performance management module

- OKR framework: Goal entity with parent-child hierarchy, weight, progress tracking
- Review cycle state machine: GOAL_SETTING → MID_YEAR → ANNUAL → CALIBRATION → PUBLISHED
- 360-degree feedback: peer nomination, manager assignment, subordinate review forms
- Rating normalization: bell-curve engine with configurable distribution percentages
- Succession planning: high-potential flag, readiness level, replacement candidates

Day 11 – React 19 frontend bootstrap

- Vite 6 React 19 TypeScript project: shadcn/ui, Tailwind 4, Zustand, TanStack Query v5
- Protected routes with role-based rendering: useAuth hook + RoleGuard component
- Layout system: sidebar navigation, breadcrumbs, notification bell, dark mode toggle
- API client: Axios interceptor for JWT refresh, global error boundary, retry logic
- Storybook setup: component library documentation for HR-specific components

Day 12 – Employee & Manager dashboards

- Employee self-service: profile editor, leave application form, payslip download grid, goals
- Manager dashboard: team attendance heatmap (D3), leave calendar, performance overview
- TanStack Query: optimistic updates for leave requests, infinite scroll for employee list
- Recharts: headcount pyramid, monthly attrition trend, department cost breakdown
- WebSocket consumer: real-time attendance status, notification toasts (Sonner)

Day 13 – Admin dashboard & HR analytics

- HR Admin dashboard: workforce KPIs, headcount waterfall, absenteeism heatmap
- Payroll admin: run initiation, approval workflow UI, payroll cost trend chart
- Org chart: React Flow (node-based, draggable), with hierarchy level filtering
- Report builder foundation: field selector component, filter builder, preview table
- Dark mode: CSS variables + Tailwind dark: prefix; system preference detection

Day 14 – Week 2 checkpoint

- Deliverables: payroll engine (PDF + journal), performance module, full React frontend connected to API
- End-to-end test: employee onboarding → attendance → payroll run → payslip download
- Lighthouse score: Performance ≥ 85, Accessibility ≥ 90, Best Practices ≥ 90
- Backend integration test suite: Testcontainers + RestAssured (60% coverage)
- Week 3 plan confirmed: AI module scope, Kafka topology, notification service

Week 3 – AI insights, real-time features & notification engine**Day 15 – Spring AI integration & attrition model**

- Spring AI 1.0 setup: OpenAI ChatClient + Ollama fallback for on-prem deployments
- Attrition prediction: Python FastAPI sidecar with Random Forest model (scikit-learn)
- Feature engineering: tenure, performance trend, salary change, absence frequency, promotion lag
- SHAP explainability: per-employee attrition drivers returned as structured JSON
- Model serving: MLflow tracking + versioning; Spring AI REST client to Python sidecar

Day 16 – Skill gap analysis & engagement scoring

- Skill extraction: NLP on employee profiles vs job descriptions (HuggingFace sentence-transformers)
- Skill gap matrix: department-level and individual-level gap heatmap with learning recommendations
- Engagement scoring: composite score from survey responses, check-in frequency, goal progress, peer recognition
- HR policy Q&A chatbot: Spring AI + RAG (Elasticsearch as vector store), policy document ingestion
- AI dashboard widget: top 10 at-risk employees, skill gaps by department, engagement trend

Day 17 – Real-time WebSocket features

- STOMP WebSocket broker: Spring WebSocket with SockJS fallback for cross-browser support
- Real-time attendance board: live clock-in/out events pushed to manager dashboard
- Notification centre: in-app toast + badge count update via WebSocket
- Payroll run progress: real-time progress bar (0–100%) as virtual threads process employees
- Presence indicator: online/offline status for HR team collaboration view

Day 18 – Notification & communication engine

- Kafka consumer group: notification-service subscribes to HR events (leave, payroll, appraisal, alert)
- Email templates: Thymeleaf HTML templates for each notification type; AWS SES batch API
- SMS channel: Twilio API integration with Textlocal fallback; OTP delivery for MFA
- Push notifications: Firebase Cloud Messaging (FCM) for PWA and mobile apps
- Dead-letter queue: failed notifications retried 3x then moved to DLQ with alerting

Day 19 – Compliance, audit & GDPR features

- Immutable audit log: append-only PostgreSQL table with SHA-256 hash chaining per row
- GDPR endpoints: data subject access request, right to erasure (soft-delete cascade), consent management
- Statutory reports: EPF ECR XML generator, ESI Excel format, Form 16 PDF batch
- Data retention policy: Quartz scheduler auto-archives records past retention period
- SonarQube quality gate: OWASP security hotspots review, code smell resolution

Day 20 – Recruitment module & self-service features

- Job requisition workflow: create → approve → publish → ATS webhook integration
- Candidate pipeline: status tracking, interview schedule (Google Calendar API webhook)
- Digital onboarding kit: auto-generated checklist, DocuSign e-signature API integration
- Employee self-service: expense claims, travel requests, IT asset requests (CRUD + workflow)
- Knowledge base: HR policy documents indexed in Elasticsearch, AI-powered search

Day 21 – Week 3 checkpoint

- Deliverables: AI module (attrition + skill gap + chatbot), WebSocket real-time features, notification engine, compliance/audit
- AI accuracy validation: hold-out set evaluation; attrition model AUC ≥ 0.82 on sample data
- Notification smoke test: leave approval → Kafka → email + SMS + in-app within 30s
- Security scan: OWASP ZAP baseline scan on staging environment; fix all High severity findings
- Performance test: 500 concurrent WebSocket connections stable for 10 minutes

Week 4 – Deployment, observability, security hardening & delivery

Day 22 – Docker multi-stage builds & Compose

- Multi-stage Dockerfiles: Maven build stage → Distroless Java 21 runtime (non-root user)
- Frontend Dockerfile: Node build stage → Nginx 1.27 static serving with gzip + cache headers
- docker-compose.prod.yml: all services with health checks, resource limits, named volumes
- Container security: CAP_DROP ALL, read-only root filesystem, seccomp profile
- Image scanning: Trivy in CI; fail build on CRITICAL CVEs; Snyk for dependency audit

Day 23 – Kubernetes manifests & Helm chart

- Helm chart structure: values.yaml + templates (Deployment, Service, Ingress, HPA, PDB, NetworkPolicy)
- Kubernetes secrets: Sealed Secrets (Bitnami) for all credentials; ConfigMap for non-sensitive config
- HPA: scale auth/employee/payroll services on CPU > 70%; Payroll engine scales on queue depth
- Istio service mesh: mTLS between all services; VirtualService for canary deployments
- AWS EKS: cluster config via Terraform; node groups with spot instances (non-prod) + on-demand (prod)

Day 24 – GitHub Actions CI/CD pipeline

- Workflow matrix: Java lint (Checkstyle) → unit tests (JUnit 5) → integration tests (Testcontainers) → Jacoco coverage gate (≥ 70%) → Docker build → push to ECR
- Frontend pipeline: ESLint → TypeScript check → Vitest → Playwright E2E → Lighthouse CI → Docker build
- ArgoCD: staging auto-sync on main branch; production requires manual approval + Slack notification
- Semantic versioning: automated release notes from conventional commits (semantic-release)
- SBOM generation: CycloneDX for software bill of materials; published to release assets

Day 25 – Cloud deployment (AWS EKS)

- AWS EKS deployment: apply Helm chart to staging, smoke test, promote to production
- RDS Aurora PostgreSQL Serverless v2: automated backups (7-day), multi-AZ, performance insights
- ElastiCache Redis: cluster mode, automatic failover, TLS in-transit
- AWS SES: DKIM/SPF/DMARC configuration; sending quota increase request; bounce handling
- CloudFront: React SPA distribution with custom domain + ACM certificate (Let's Encrypt fallback)

Day 26 – Observability & monitoring stack

- Prometheus: scrape configs for all Spring Boot /actuator/prometheus endpoints + JVM metrics
- Grafana dashboards: API latency heatmap, JVM heap, Kafka consumer lag, DB connection pool, payroll run progress
- Loki: structured JSON log aggregation from all pods; log-based alerts for ERROR patterns
- Jaeger: distributed traces via OpenTelemetry SDK; trace sampling 100% errors / 10% success
- PagerDuty: on-call routing for P1 incidents; SLA breach alerts; payroll run failure escalation

Day 27 – Load testing, security hardening & QA

- k6 load test: 10,000 concurrent virtual users for 15 minutes; validate P95 < 300ms; no errors
- OWASP ZAP full scan: active scan on staging; fix all High/Critical findings before production sign-off
- Penetration test checklist: SQL injection, XSS, IDOR, JWT manipulation, SSRF, mass assignment
- GDPR audit: verify PII masking, consent flows, data export, erasure cascade works end-to-end
- Cross-browser QA: Chrome, Firefox, Safari, Edge; mobile responsive at 375px / 768px / 1440px

Day 28 – Final delivery, demo video & documentation

- Record 5–7 min scenario-based demo: onboarding → attendance → payroll run → AI insights → notification
- README.md: monorepo structure, docker-compose local setup, environment variables, architecture screenshots
- Postman collection: all 60+ endpoints with example payloads, environment variables, automated tests
- Model card: attrition model training data, AUC, feature importance, bias evaluation, intended use
- Submission package: PDF report + GitHub repo + live HTTPS demo URL + demo video link

6

Security, Compliance & Data Privacy

Domain	Controls Implemented	Standard / Tool
Authentication	Keycloak OIDC/SAML; JWT RS256; Argon2id hashing; MFA (TOTP); session blacklist in Redis	NIST 800-63B / Zero Trust
Authorisation	Spring Security RBAC + ABAC; @PreAuthorize on every endpoint; tenant isolation via RLS	NIST 800-53 AC-6
API Security	CORS allowlist; CSRF protection; rate limiting (Redis sliding window); input validation (Bean Validation)	OWASP API Security Top 10
Data Encryption	TLS 1.3 in transit; AES-256 at rest (RDS + S3); field-level encryption for PAN/Aadhaar	FIPS 140-2 / PCI DSS
Injection Prevention	Parameterised JPQL + native queries via Hibernate; no string concatenation in queries; XSS via DOMPurify	OWASP A03:2021
Secrets Management	AWS Secrets Manager + Kubernetes Sealed Secrets; zero hardcoded credentials; Vault for dynamic secrets	CIS K8s Benchmark
Audit Logging	Immutable append-only audit table; SHA-256 hash chaining; all PII mutations logged with before/after JSON	SOC 2 CC7.2
GDPR Compliance	DSR endpoints (access/erasure/portability); consent management; data retention schedules; DPA agreements	GDPR Art. 17, 20, 25
Dependency Security	Snyk SCA in CI; Trivy container scan; OWASP Dependency-Check; Renovate bot for auto-updates	CIS Controls v8
Code Quality	SonarQube quality gate (0 critical issues); Checkstyle + SpotBugs in CI; mandatory code review	ISO 27001 A.14.2.2

7

Technical Highlights & Architecture Patterns

Pattern / Practice	Implementation in NexusHR	Business Benefit
Domain-Driven Design (DDD)	Each service owns its bounded context (Payroll, Attendance, Performance); aggregates enforce invariants; no cross-domain DB joins	Loose coupling; independent deployability; clear ownership
CQRS (lightweight)	Read models (PostgreSQL views + Elasticsearch) separate from write models for dashboards and reports	Optimised query performance; no OLTP/OLAP contention
Event Sourcing (audit log)	Immutable audit_log table captures every state change with before/after JSON + hash chaining	Full audit trail; GDPR compliance; time-travel debugging
Saga Pattern	Payroll run: multi-step saga (calculate → validate → approve → post → notify) with compensating transactions on failure	Zero inconsistent payroll states; automatic rollback on failure
Java Virtual Threads	Payroll bulk processing: 10,000 employee calculations parallelised with virtual threads (Project Loom)	10x throughput vs platform threads; no reactive complexity
Outbox Pattern	Domain events written to outbox table in same transaction as business data; Kafka producer reads outbox	Guaranteed event delivery; no dual-write problem
Circuit Breaker	Resilience4j on AI sidecar calls and third-party APIs (Twilio, DocuSign); fallback responses defined	Prevents cascade failures; graceful degradation
Blue-Green Deployment	Kubernetes: two identical environments; Istio traffic shift; instant rollback by flipping VirtualService weight	Zero-downtime deployments; instant rollback < 30 seconds

8 Evaluation Criteria (100-Point Scale)

Category	Points	Focus Areas
Innovation & Problem Solving	15	AI attrition prediction originality, thoughtful UX design, multi-tenant architecture decisions
Technical Depth & Best Practices	25	DDD, CQRS, Saga, virtual threads, security hardening, OWASP compliance, design patterns
Functionality & User Experience	20	All F-01–F-10 met, responsive design, accessibility (WCAG 2.1 AA), dark mode, intuitive UX
Documentation Quality	20	Professional report, OpenAPI spec, ADRs, model card, README depth and architecture diagrams
Deployment & Reliability	10	Stable HTTPS demo, CI/CD pipeline, Kubernetes HPA, load test results, monitoring dashboard
Presentation & Polish	10	Demo video quality, Lighthouse scores, dashboard visual design, storytelling effectiveness
TOTAL	100	

9 Submission Guidelines

#	Deliverable	Format / Location	Required?	Weight
1	Project Report (this document)	1 PDF, A4, 10–18 pages	Yes	25%
2	Live Public Demo URL	HTTPS, no mandatory login (demo credentials on page)	Yes	30%
3	GitHub Repository	Public monorepo with clear module folders	Yes	20%
4	README.md	Setup guide, architecture diagrams, env vars, screenshots, video	Yes	15%
5	Demo Video	5–7 min, YouTube Unlisted / Loom / Google Drive	Yes	10%

Naming convention: YourName_NexusHR_AmdoxJFS_April2026.pdf / .zip

Submission: Via the Amdox Technologies project dashboard only — no email submissions. Late submissions are automatically disqualified from stipend consideration.

File limits: ZIP ≤ 500 MB | PDFs ≤ 12 MB | No private API keys required for judges to run demo.

Stand-out practices (highly recommended)

- Include Lighthouse scores, k6 load test report, and OWASP ZAP scan summary in the report
- Record a live multi-user session in the demo video (manager approving leave, payroll run, AI attrition view)
- Demonstrate the Kafka event flow visually: leave approved → notification toast → email received

- Show the CI/CD pipeline run end-to-end in the video with the GitHub Actions log
- Add Architecture Decision Records (ADRs) to the GitHub repo for the top 5 design decisions
- End the report with a Personal Reflection: key learnings, Java 21 virtual thread challenges, future roadmap

Crafted with precision and modern engineering principles

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